

CASINO, Australia

WMO: 945730

Lat: 28.878S

Lon: 153.052E

Elev: 22

StdP: 101.06

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	5.0	-2.8	3.0	17.0	-0.9	3.5	14.9	8.9	19.6	7.9	18.7	2.2	280	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	35.2	22.0	33.2	22.0	31.4	21.4	24.4	31.3	23.7	30.0	23.0	28.7	5.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.3	26.8	21.9	16.6	26.1	21.3	16.0	25.6	73.9	31.6	71.0	30.1	68.4	28.8	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.6	0.9	39.4	1.5	2.5	-0.2	41.2	-1.0	42.7	-1.9	44.1	-2.9	45.9		
(5)			0.1	25.5	1.5	0.8	-1.0	26.1	-1.8	26.5	-2.7	27.0	-3.7	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	24.5	24.0	22.8	20.0	16.6	14.4	13.7	14.6	17.8	20.1	22.0	23.6		
	DBStd	4.52	2.49	2.25	2.01	1.90	2.08	2.16	2.20	2.35	2.67	2.83	2.95	2.80		
	HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0		
	HDD18.3	506	0	0	0	6	61	119	146	118	42	13	2	1		
	CDD10.0	3462	450	393	397	299	204	132	114	144	233	313	359	423		
	CDD18.3	926	192	159	138	55	7	1	1	4	25	68	111	165		
	CDH23.3	7352	1599	1178	828	286	62	7	15	76	296	659	958	1389		
	CDH26.7	2593	638	422	236	45	5	1	1	12	75	224	374	562		
(14) Wind	WSAvg	2.9	3.2	3.1	2.7	2.4	2.3	2.6	2.6	2.8	3.0	3.2	3.3	3.2		
(15) Precipitation	PrecAvg	1124	124	166	145	90	96	61	59	38	30	71	85	139		
	PrecMax	1734	302	481	429	485	432	396	335	174	113	379	208	405		
	PrecMin	633	9	21	10	10	11	8	0	0	0	7	21	14		
	PrecStd	285	76	94	105	101	94	75	72	36	27	67	47	77		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	36.1	34.4	30.0	27.2	24.1	25.2	28.1	31.8	34.6	36.2	37.7		
		MCWB	23.5	23.8	22.5	20.2	17.9	16.4	15.4	15.6	17.7	19.1	20.4	21.9		
	2%	DB	34.4	32.9	30.9	28.0	25.0	22.4	22.8	25.3	28.7	31.3	33.3	34.4		
		MCWB	23.0	23.7	22.1	19.6	16.9	15.4	14.7	15.0	17.2	18.4	20.7	22.0		
	5%	DB	32.1	30.9	29.1	26.3	23.4	21.0	21.1	23.2	26.7	29.1	30.7	31.8		
		MCWB	22.6	22.4	21.5	19.1	16.4	14.7	13.9	14.1	16.6	18.2	20.4	21.5		
	10%	DB	30.1	29.2	27.5	24.9	22.0	19.5	19.5	21.3	24.6	27.0	28.4	29.7		
		MCWB	22.4	22.0	21.0	18.7	15.9	14.2	13.3	13.4	15.9	17.9	19.9	21.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.4	24.2	21.9	20.0	18.2	18.2	17.8	19.9	22.0	23.4	24.7		
		MCDB	33.0	32.5	29.9	27.2	24.2	21.8	22.6	24.5	28.5	29.3	31.9	32.8		
	2%	WB	24.3	24.4	23.2	20.9	18.6	17.0	16.6	16.5	18.7	20.6	22.2	23.7		
		MCDB	31.1	30.8	28.3	25.6	23.2	20.1	20.1	21.7	25.8	27.0	29.4	31.2		
	5%	WB	23.6	23.6	22.5	20.1	17.7	16.2	15.3	15.5	17.9	19.7	21.4	22.8		
		MCDB	29.8	29.1	27.1	24.4	21.6	19.1	19.0	20.3	24.2	25.6	27.9	29.2		
	10%	WB	22.9	22.8	21.8	19.4	16.9	15.3	14.3	14.6	17.1	18.8	20.7	22.1		
		MCDB	28.4	27.3	26.0	23.5	20.4	18.2	17.8	19.4	22.8	24.7	26.5	27.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	9.7	9.4	10.3	11.8	11.2	12.8	14.3	13.8	12.8	11.8	11.1		
		MCDBR	14.4	13.1	12.4	12.8	13.7	13.0	15.2	17.5	16.9	16.9	15.7	15.1		
	5% WB	MCWBR	5.2	5.0	4.9	5.6	6.7	7.1	7.9	8.4	7.0	6.5	5.7	5.4		
		MCWBR	11.8	11.3	10.2	10.5	11.0	10.1	11.3	13.6	14.0	13.3	12.7	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.389	0.374	0.356	0.343	0.313	0.307	0.299	0.312	0.354	0.381	0.386	0.396		
	taud		2.475	2.530	2.575	2.582	2.626	2.635	2.641	2.583	2.459	2.415	2.447	2.441		
	Ebn at Noon		953	950	936	901	886	869	894	920	922	932	949	949		
	Edh at Noon		118	109	98	88	76	72	74	86	108	120	121	123		
(44) All-Sky Solar Radiation	RadAvg		6.46	5.86	4.98	4.26	3.53	2.92	3.38	4.28	5.31	5.92	6.38	6.44		
	RadStd		0.54	0.49	0.29	0.20	0.23	0.19	0.26	0.33	0.43	0.55	0.57	0.65		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.60	N/A	N/A	+1.68	N/A	N/A	N/A	-83	+262	+191
(47) Regional (21 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-58	+169	+104

Nomenclature: See separate page